Module 2 Assignment – Student Record Manager

Java Program

```
import java.io.*;
import java.util.*;
import java.util.regex.Pattern;

public class StudentRecordManager {

    static final String FILE_NAME = "students.txt";

    // Validate email using Regex
    static boolean isValidEmail(String email) {
        String emailRegex = "^[A-Za-z0-9+_.-]+@[A-Za-z0-9.-]+$";
        return Pattern.matches(emailRegex, email);
    }

    // Add Student
    static void addStudent(Scanner sc) {
        try {
            System.out.print("Enter Student ID: ");
            int id = Integer.parseInt(sc.nextLine());

            System.out.print("Enter Student Name: ");
            String name = sc.nextLine();

            System.out.print("Enter Email: ");
            String email = sc.nextLine();

            if (!isValidEmail(email)) {
                throw new IllegalArgumentException("Invalid email format!");
            }

            // Save data to file
            FileWriter fw = new FileWriter(FILE_NAME, true);
            BufferedWriter bw = new BufferedWriter(fw);

            bw.write(id + "," + name + "," + email);
            bw.newLine();

            bw.close();

            System.out.println("Student added successfully!");

        } catch (NumberFormatException e) {
            System.out.println("Invalid Student ID! Please enter a number.");
        } catch (IllegalArgumentException e) {
            System.out.println(e.getMessage());
        } catch (IOException e) {
            System.out.println("Error saving student data.");
        }
    }

    // Read Student Data
    static void readStudents() {
        try {
            File file = new File(FILE_NAME);

            if (!file.exists()) {
                System.out.println("No student records found.");
                return;
            }

            BufferedReader br = new BufferedReader(new FileReader(file));

            String line;

            System.out.println("\n--- Student Records ---");

            while ((line = br.readLine()) != null) {
                String[] data = line.split(",");

                System.out.println("ID: " + data[0]);
                System.out.println("Name: " + data[1]);
                System.out.println("Email: " + data[2]);
                System.out.println("----------------------");
            }
            br.close();
```

```
        } catch (IOException e) {
            System.out.println("Error reading student data.");
        }
    }

    // Main Method
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        while (true) {

            System.out.println("\n===== Student Record Manager =====");
            System.out.println("1. Add Student");
            System.out.println("2. Read Student Data");
            System.out.println("3. Exit");

            try {
                System.out.print("Enter your choice: ");
                int choice = Integer.parseInt(sc.nextLine());

                switch (choice) {

                    case 1:
                        addStudent(sc);
                        break;

                    case 2:
                        readStudents();
                        break;

                    case 3:
                        System.out.println("Thank you!");
                        sc.close();
                        return;

                    default:
                        System.out.println("Invalid choice! Please select 1, 2 or 3.");
                }

            } catch (NumberFormatException e) {
                System.out.println("Invalid input! Please enter a number.");
            }
        }
    }
}
```